Vishnu Datta

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Education

Indian Institute of Information Technology, Sri City **Aug 2023 – May 2027**
Bachelor of Technology in Computer Science and Engineering *Sri City, India*

Experience

Soundverse AI **Apr 2026 – Oct 2026**
Software Development Engineer Intern *Remote*

- Scaled a **Whisper** transcription and speaker-diarization pipeline from **25 MB** to **500 MB** inputs with silence-aware chunking, parallel chunk transcription and live progress over **Server-Sent Events**
- Built core services of the **Music Distribution** pipeline that lets Artists release songs to Spotify, Apple Music and YouTube: with media conversion, metadata mapping, album submission and per-store status tracking
- Designed and built a fail-closed **quality gate** of 7 weighted checks (loudness, lyrics, artwork, ISRC conflicts, AI provenance, copyright, **AWS Rekognition** moderation) that blocks bad releases before any store
- Built the **Media Converter** end to end for audio, video and image files: background jobs with live **Socket.IO** progress, per-plan rate limits in **Redis** and token billing; fixed a proxy that stalled large files for 40–200 s
- Co-built a video playback overhaul moving delivery from MP4 to **HLS over a CDN**, benchmarking seeks **7.6x faster** and first byte **6x faster**; owned link minting with one indexed ownership lookup and background rendition prep
- Designed a cross-channel **notification system**: one stored record per event feeds the in-app bell, **Firebase** push and **Amazon SES** email, with a scheduler that holds the email and cancels it once the user reads it anywhere
- Worked across **API design** (gRPC/Protobuf, REST), **PostgreSQL** schema design and third-party **API integrations**: Didit identity verification (webhooks, audit trail), Sign in with Apple and Google Play Billing

Projects

Audira AI | *Next.js, TypeScript, Python, FastAPI, MongoDB, WebSockets, Judge0* | GitHub | Live Demo

- Built an **AI technical interviewer** that reads each candidate's résumé against the job description, then runs coding, project and fundamentals rounds, probing vague answers and digging into claims like a human interviewer
- Made LLM-written coding problems **trustworthy**: expected outputs come from executing a hidden reference solution, broken problems are regenerated, and submissions run sandboxed on **Judge0** in **13 languages**
- Generated recruiter reports that tie every score to the candidate's own words, validated by an **eval harness** that must separate a genuine candidate from a bluffer on one résumé; runs on multi-provider LLM failover

CodeRep | *TypeScript, Next.js, PostgreSQL, Prisma, Clerk, Chrome Extension (MV3)* | GitHub | Live Demo

- Launched a spaced-repetition system for LeetCode, a **Next.js** web app plus **Chrome Extension** used by **20+ daily active users**, that catches accepted submissions with **99% accuracy** via a **MutationObserver**
- Designed an adaptive review scheduler (reset on failure, 1.2x–4x interval growth, 60-day mastery cap) on **PostgreSQL** and Prisma, with Clerk OAuth for the web app and scoped API tokens for the extension

TaskLane | *Django REST Framework, PostgreSQL, Redis, Celery, Next.js, TypeScript, JWT* | GitHub

- Built a multi-tenant, Jira-style work tracker where teams run projects on a drag-and-drop board, with **role-based access** (Owner, Manager, Member) enforced server-side and an **append-only audit log** of every task change
- Kept the analytics dashboard at a **constant 4 SQL queries** at any data size with in-database aggregation and per-user **Redis** caching, and moved email and hourly overdue sweeps off the request path with **Celery**

Achievements

Regional Winner, GDG Bit N Build International Hackathon; qualified for the International Grand Finals in Mumbai

Technical Skills

Languages: Python, TypeScript, JavaScript (ES6+), Java, C, SQL, HTML, CSS
Backend: FastAPI, Node.js, Express.js, Django REST Framework, gRPC, Protobuf, REST, WebSockets, JWT
Frontend: React.js, Next.js (App Router), Tailwind CSS, Vite, Chrome Extensions (Manifest V3)
Databases & Cloud: PostgreSQL, MongoDB, Redis, AWS (Rekognition, Comprehend, SES), Firebase, Azure, Docker
AI & Fundamentals: LLM Integration, OpenAI Whisper, Gemini API, FFmpeg, DSA, OOP, DBMS, System Design